

6. Banking

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Road Map

Asset Transformation

Credit Risk

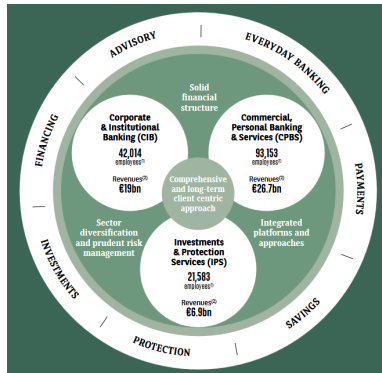
Interest Rate Risk

Liquidity Risk

Banks

- Large banks are active in
 - ▶ Commercial banking: take deposits, make loans
 - ▶ Investment banking: underwriting, M&A, trading
 - ▶ Asset management

- Ex.: BNP Paribas



Commercial Banking

- Today: focus on commercial banking



What are the economic functions of commercial banks – why don't households lend directly to firms and to other households?

Asset Transformation

Take deposits + Make loans = Asset transformation

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1. Liquidity transformation

- ▶ Deposits are liquid: they are demandable
- ▶ Loans are illiquid: banks cannot call loans; secondary markets for loans exist but are illiquid

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2. Maturity transformation

- ▶ Borrow short-term: deposits, certificates of deposits, repurchase agreements (repos), interbank loans
- ▶ Lend long-term: business loans, mortgages, government bonds

Asset Transformation

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3. Credit risk transformation

- ▶ Deposits are safe
- ▶ Loans are risky

Asset Transformation

- Remark: Liquidity and maturity are related characteristics (that deposits are demandable make them both liquid and short-term) but different



Can you name assets that are liquid and long-term?

Diversification

- Liquidity transformation and credit risk transformation are achieved through diversification
 - ▶ Not all depositors withdraw their deposits at the same time
 - ▶ Not all loans default
- Maturity transformation cannot be diversified away
 - ▶ Interest rate risk is systematic

Risks

Transforming illiquid, long-term, risky assets into liquid, short-term, safe liabilities exposes banks to:

1. Liquidity risk
2. Interest rate risk
3. Credit risk

Road Map

Asset Transformation

Credit Risk

Interest Rate Risk

Liquidity Risk

Credit Risk

How do banks manage credit risk?

- **Limit individual risk**

- ▶ Screening (e.g., credit scoring as studied in class 2)
- ▶ Collateral: Loans may be secured by a specific asset (equipment, real estate, account receivables, entrepreneurs' personal assets)
- ▶ Monitoring: Loans may include covenants (e.g., limits on debt-to-EBITDA, on dividends)

- **Diversify risk:** Hold loans in different sectors, different geographies, etc.

- **Sell risk**

- ▶ Securitization: Sell part of the loans cash flow
- ▶ Hedge using credit derivatives such as credit default swaps (uncommon)

Banks must hold equity to absorb losses due to remaining credit risk

Road Map

Asset Transformation

Credit Risk

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Liquidity Risk

Interest Rate Risk

By borrowing short-term and lending long-term (at fixed rate), banks:

- Earn positive net interest income because long rates are usually higher than short rates (i.e., the **term premium** is positive)
- Are exposed to the risk of an increase in interest rate

Interest Rate Risk: Example

- Deposits 100: demandable, interest rate 1%/year
- Loans 100: maturity 5 years, interest rate 2%/year (fixed rate)
- Cash flow if **no change** in interest rates (ignoring compounding)

	year 0	year 5
deposits	100	-105
loans	-100	110
total		+5

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- If the yield curve **shifts up by 2%**

	year 0	year 5
deposits	100	-115
loans	-100	110
total		-5

Duration

Refresher from Financial Markets Course

- The loss of 10 can be also calculated using the duration formula:

$$(\Delta \text{Value}) \simeq - (\Delta \text{Interest rate}) \times (\text{Duration in years})$$

- Yield curve shifts up by 2%
- Duration of assets: 5 years
Duration of liabilities: 0

$$\Rightarrow \Delta \text{Value of assets} \simeq -2\% \times 5 = -10\%$$
$$\Delta \text{Value of liabilities} \simeq -2\% \times 0 = 0$$

Interest Rate Risk

How do banks manage interest rate risk?

- **Asset Liability Management:** Manage the mismatch between assets duration and liabilities duration
 - ▶ Borrow longer term
 - ▶ Lend shorter term
 - ▶ Lend at variable rate
 - ▶ Hedge using interest rate derivatives
- But reducing the duration mismatch implies giving up on the term premium

Banks hold equity to absorb losses due to remaining interest rate risk



SVB is largest bank failure since 2008 financial crisis

March 11, 2023 6:15 AM GMT+1 · Updated March 11, 2023



Silicon Valley Bank: Rising interest rates will uncover more ticking bombs

12 March 2023

U.S. Banking System, March 2023

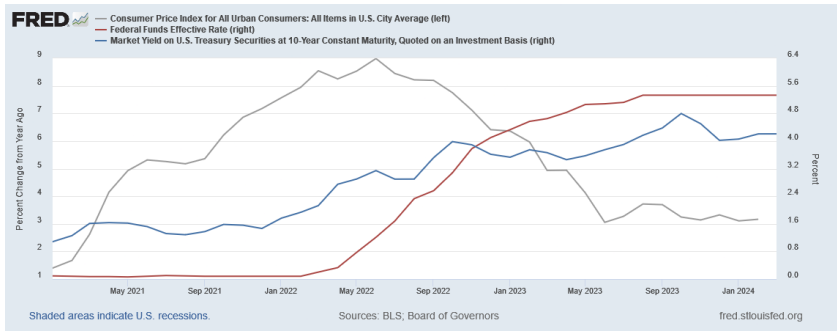
Bank Asset Composition – 2022Q1

Total Asset \$	24T
Number of Banks	4844
(Percentage of Asset)	
Cash	14.1
Security	25.2
Treasury	6.1
RMBS	12.1
CMBS	2.3
ABS	2.7
Other Security	2.1
Total Loan	46.6
Real Estate Loan	21.9
Residential Mortgage	10.6
Commercial Mortgage	2.2
Other Real Estate Loan	9.1
Agricultural Loan	0.3
Commercial & Industrial Loan	9
Consumer Loan	7.7
Loan to Non-Depository	2.8
Fed Funds Sold	0.1
Reverse Repo	1.2

Bank Liability Composition – 2022Q1

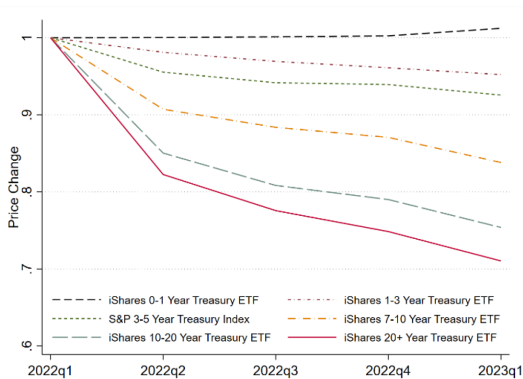
Total Liability	90.5
Domestic Deposit	76.6
Insured Deposit	41.1
Uninsured Deposit	37.4
Uninsured Time Deposits	1.8
Uninsured Long-Term Time Deposits	0.4
Uninsured Short-Term Time Deposits	1.3
Foreign Deposit	6.5
Fed Fund Purchase	0.1
Repo	0.6
Other Liability	2.3
Total Equity	9.5
Common Stock	0.2
Preferred Stock	0.1
Retained Earning	4

U.S. March 2023



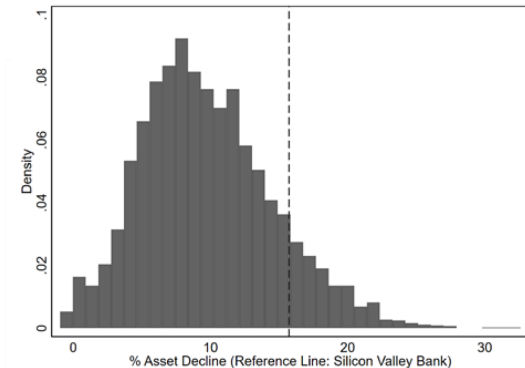
Source: <https://fred.stlouisfed.org/>

U.S. Banking System, March 2023



(c) Treasury by Maturity

Distribution of Change in Asset Value (“Marking to Market”)



Source: Jiang, Matvos, Piskorski and Seru, “Monetary Tightening and U.S. Bank Fragility in 2023: Mark-to-Market Losses and Uninsured Depositor Runs?” Journal of Financial Economics, 2024 [\[pdf\]](#)

Interest Rate Risk



Most U.S. banks did not fail in 2023 despite large losses, sometimes larger than equity. Why not?

Interest Rate Risk: Sticky Deposit Rates



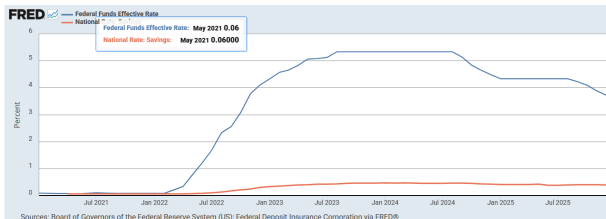
Most U.S. banks did not fail in 2023 despite large losses, sometimes larger than equity. Why not?

- Deposits do not fully reprice
 - ▶ Back to our example: deposit rate 1%, 5-year loan fixed rate 2%
 - ▶ Suppose yield curve shifts up by 2%

... but deposit rate increases by 1% only

	year 0	year 5
deposits	100	-115 -110
loans	-100	110
total		-5 0

Interest Rate Risk: Sticky Deposit Rates



- The **deposit beta**, defined as the sensitivity of deposits rates to changes in the short rate, is far below one
- Due to depositors “sleepiness” giving banks market power in deposits rate setting

Road Map

Asset Transformation

Credit Risk

Interest Rate Risk

Liquidity Risk

Liquidity Risk — Game

- Players
 - ▶ 1 bank: Johan
 - ▶ N short-term creditors: You and the N-1 other students in the room
- You hold 100 worth of debt on the bank expiring now. Choose btw:
 - ▶ Roll-over at 2% interest rate → the bank owes you 102 at maturity
 - ▶ Stop lending → the bank pays you 100 now

- Bank's balance sheet:

Assets	Liabilities
Loans: $N \times 100$	Short-term debt: $N \times 100$

- Bank's assets payoff to the bank (per \$1 of book value today):
 - ▶ 1.05 at maturity if held until maturity
 - ▶ 1 now if liquidated now

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Decide whether to roll-over or stop lending
<https://forms.gle/jXmcEkGFKT94fRnw5>

Illiquid Bank's Assets

- Now, the bank's assets are **illiquid**. They pay off:

1.05 at maturity if held until maturity

0.80 now if liquidated now

- The rest is the same

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[spreadsheet]

Bank Runs

- Best response depends on expectations of what other depositors do
 - ▶ If other depositors roll over $\Rightarrow$ Should roll over
 - ▶ If other depositors stop lending $\Rightarrow$ Should stop lending

$\Rightarrow$ Two possible (self-fulfilling) equilibria

- ▶ Business as usual: all depositors roll over
- ▶ “Bank run”: all depositors stop lending



How to avoid bank runs?

Deposit Insurance

- Deposit insurance: Government commits to repay depositors if the bank is insolvent $\Rightarrow$ Depositors have no incentives to run $\Rightarrow$ Self-fulfilling
- Costless for the government since run does not actually occur
- Deposits are usually insured up to a maximum amount (France €100k, US \$250k)

Silicon Valley Bank, March 2023

- 90% of SVB's deposits were uninsured



Run Risk Beyond Banks

- Run risk exists for financial intermediaries with **short-term debt liabilities** + **illiquid assets**
- Are equity mutual funds exposed to runs?

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Run Risk Beyond Banks

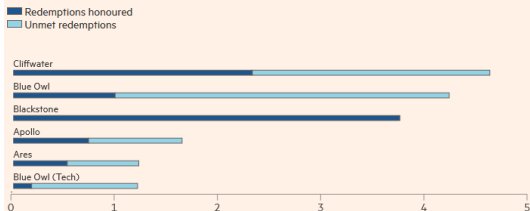
- Run risk exists for financial intermediaries with **short-term debt liabilities** + **illiquid assets**
- Are equity mutual funds exposed to runs?
- Are private equity funds exposed to runs?
- High demand for private equity and private debt funds led private capital firms to create such funds for individuals with redeemable shares

Investors sought to pull \$20bn from private credit funds in first quarter

Big groups including Apollo, Ares and Blackstone were hit with redemption requests at start of 2026

Published APR 9 2026

Quarterly redemptions fulfilled and left unmet (\$bn)



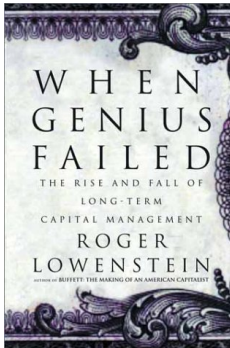
Source: Company filings, FT research

FINANCIAL TIMES

Run Risk Beyond Banks

- Hedge funds and investment banks holding illiquid assets financed with short-term debt

LTCM failure in 1998



Lehman Brothers bankruptcy in 2008
triggering the Great Financial Crisis



Theory of Runs

- The game you played is based on the Diamond-Dybvig model of bank runs (Nobel 2022)

Fresh Money Injection — With Illiquid Assets

- Suppose creditors are uninsured and there is a run. We go back to the point in time just after creditors have run, before the bank liquidates assets at 80 cents on the dollar
- A deep pocket investor steps in and can provide fresh money to the bank. If this happens, the deep pocket investor's claim is junior to the existing creditors
- Please form groups of two persons: one is the bank, one is the deep pocket investor
- Goal of each party: maximize his or her profit
- The deep pocket investor's WACC is 0%
- You can only do financial restructuring, you cannot increase the value of assets
- Negotiations open!

Fresh Money Injection — With Impaired Assets

- Now, the bank's assets are illiquid and **impaired**: they pay off
 - ▶ **0.95** at maturity if held until maturity
 - ▶ 0.80 now if liquidated now
- **Negotiations open!**

Banks Runs — Summary

- *Definitions:*

An **illiquid** bank cannot repay short-term creditors if they don't roll over.
A liquid bank can

An **insolvent** bank cannot repay creditors even if they roll over until asset maturity. A **solvent** bank can

- When the bank is liquid and solvent:

- ▶ Only equilibrium: No-run (= first round of the run game)

Banks Runs — Summary

- When the bank is **illiquid** and **solvent**:
 - ▶ Multiple equilibria: No-run and Run (= **second round of the run game**)
 - ▶ Depositors sleepiness mitigates run risk. Digital banking amplifies run risk
 - ▶ Deposits insurance prevents run
 - ▶ Without deposits insurance, a solvent bank can raise new funds (= **first negotiation game**)
- When the bank is **illiquid** and **insolvent**:
 - ▶ Run
 - ▶ An insolvent bank can't raise new funds (= **second negotiation game**)
⇒ default (Lehman Brothers 2008) or bailout (Silicon Valley Bank 2023)
 - ▶ Deposits insurance may delay but not avoid run and failure

Can you think of a potential problem created by deposit insurance?



Conclusion

- Group assignment due today
- Final exam: May 22, 45 min, 10 MCQs, pen, paper, and calculator
- On Blackboard: 6 practice problems and 2 practice exams with solution

I hope you enjoyed the class and learnt a lot!

